a

SIZE ONLY (mask area) day 30 | CNN, whole image day 30 | CNN, cropped to organoid day 30 | CNN, cropped + brightened day 30

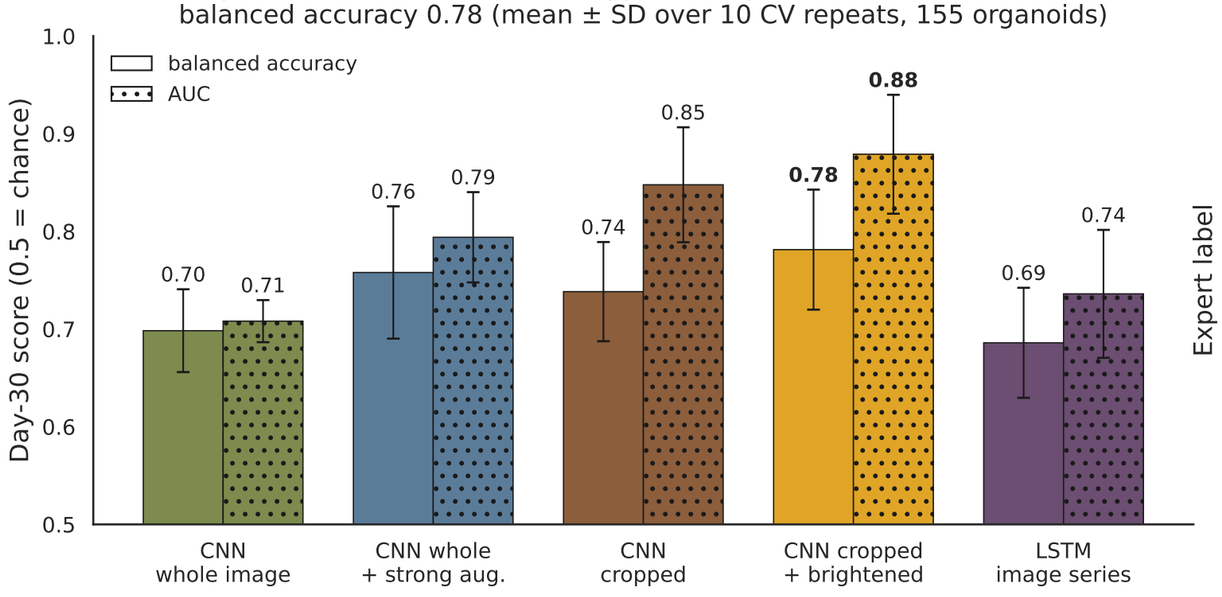
LSTM: the whole image series — day 3 | day 10 | day 17 | day 24 | day 30

BA1 1_A7 accepted (5/5 Acc. votes) — area = 2.57 mm²

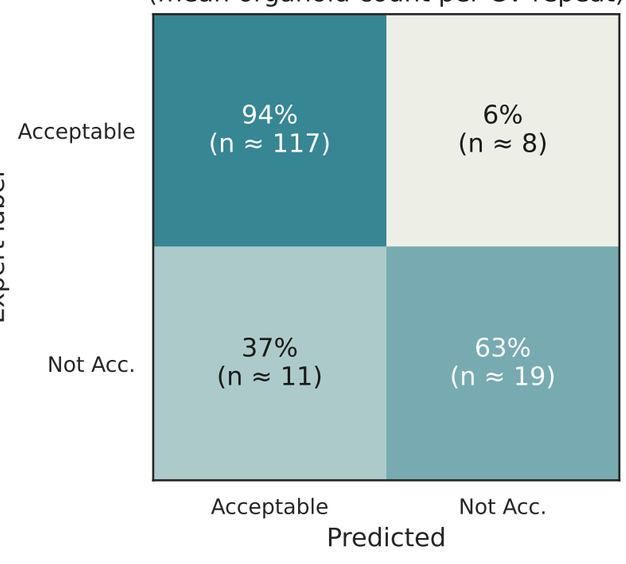
BA1 1_G1 rejected, stalled (0/5 Acc. votes) — area = 0.85 mm²

BA2 2_C6 rejected, overgrown (1/5 Acc. votes) — area = 4.09 mm²

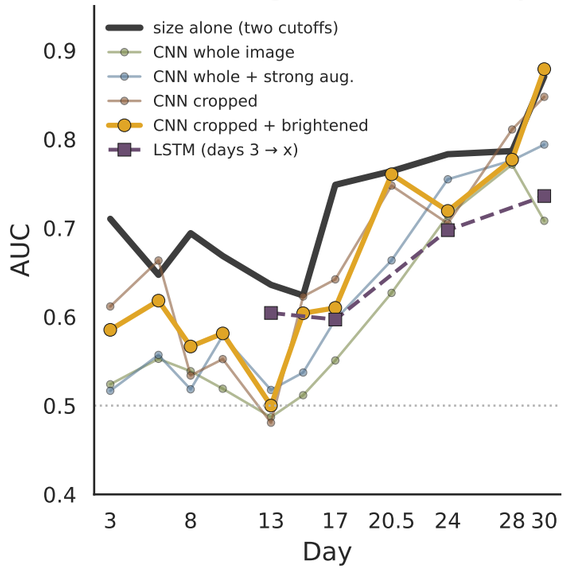
b Day 30: the best model (CNN cropped + brightened) reaches AUC 0.88, balanced accuracy 0.78 (mean ± SD over 10 CV repeats, 155 organoids)



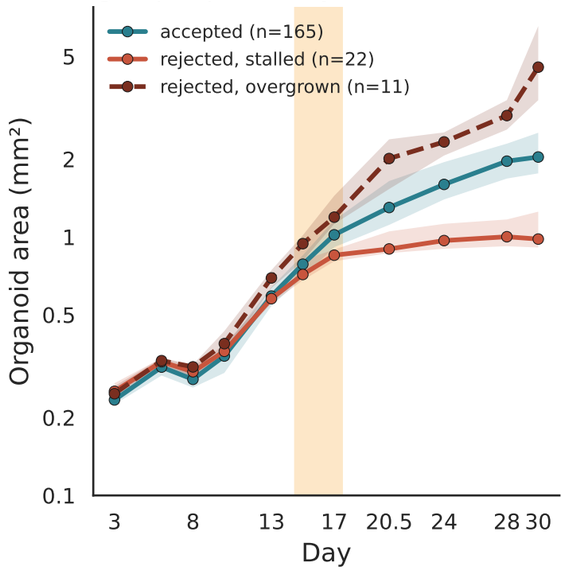
c Best model, day 30: % of each expert class (mean organoid count per CV repeat)



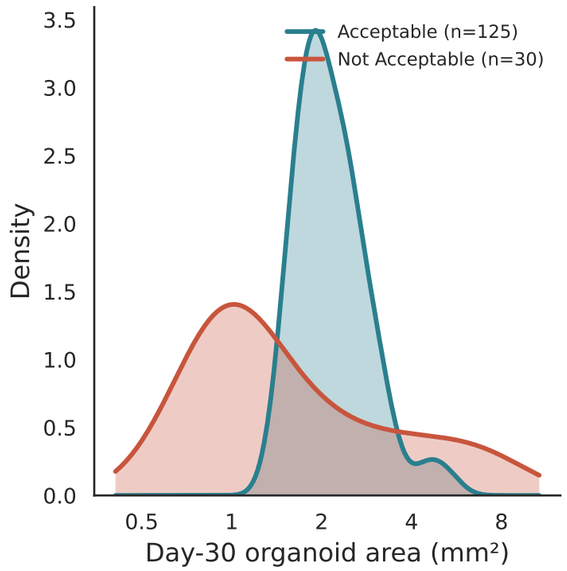
d Image models only work from ~day 20; size alone is as good or better every day



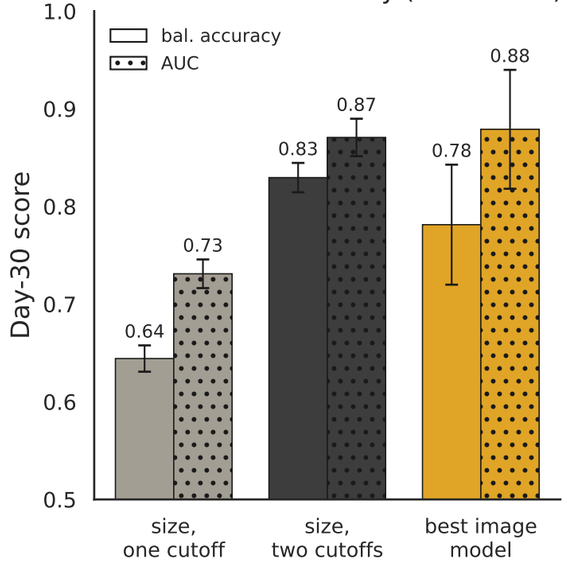
e Rejects stall or overgrow; groups split at days 15–17 (shaded)



f Not Acceptable organoids are too small or too big



g Size alone (two cutoffs) matches the best image model on AUC and beats it on balanced accuracy (same folds)



h The image series adds nothing: LSTM ≤ size + growth rate in every window

